

MIT OpenCourseWare

14.12

Economic Applications of Game Theory 14.12 — Economic Applications of Game

Theory May 2026 **EXTREMELY HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (2 points) Analyze the limitations of standard approaches to Nash Equilibrium when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
2. (5 points) Construct an original proof-level problem involving Bayesian Games for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
3. (4 points) Analyze the limitations of standard approaches to Auction Theory when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (6 points) Construct an original proof-level problem involving Bargaining for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
5. (2 points) Construct an original proof-level problem involving Mechanism Design for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
6. (5 points) Analyze the limitations of standard approaches to Signaling when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (7 points) Prove the fundamental theorem concerning Nash Equilibrium in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (3 points) Prove the fundamental theorem concerning Bayesian Games in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (2 points) Analyze the limitations of standard approaches to Auction Theory when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
10. (3 points) Construct an original proof-level problem involving Bargaining for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
11. (5 points) Prove the fundamental theorem concerning Mechanism Design in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (5 points) Analyze the limitations of standard approaches to Signaling when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
13. (6 points) Analyze the limitations of standard approaches to Nash Equilibrium when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (5 points) Prove the fundamental theorem concerning Bayesian Games in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
15. (4 points) Prove the fundamental theorem concerning Auction Theory in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
16. (2 points) Analyze the limitations of standard approaches to Bargaining when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.
17. (3 points) Construct an original proof-level problem involving Mechanism Design for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
18. (2 points) Prove the fundamental theorem concerning Signaling in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (3 points) Analyze the limitations of standard approaches to Nash Equilibrium when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (6 points) Prove the fundamental theorem concerning Bayesian Games in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
21. (5 points) Construct an original proof-level problem involving Auction Theory for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (5 points) Prove the fundamental theorem concerning Bargaining in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Prove the fundamental theorem concerning Mechanism Design in the context of 14.12 (Economic Applications of Game Theory). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (5 points) Construct an original proof-level problem involving Signaling for 14.12 (Economic Applications of Game Theory) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
25. (3 points) Analyze the limitations of standard approaches to Nash Equilibrium when applied to edge cases in 14.12. Construct explicit counterexamples showing where intuitive reasoning fails.

Solutions

Solution 1:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Nash Equilibrium. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 2:

Solution approach: First recall the definitions and key theorems related to Bayesian Games from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Auction Theory. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

Solution approach: First recall the definitions and key theorems related to Bargaining from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 5:

Solution approach: First recall the definitions and key theorems related to Mechanism Design from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Signaling. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

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Solution 10:

Solution approach: First recall the definitions and key theorems related to Bargaining from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 11:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Signaling. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Nash Equilibrium. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 15:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 16:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Bargaining. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 17:

Solution approach: First recall the definitions and key theorems related to Mechanism Design from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 18:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

The edge case analysis reveals the subtle assumptions underlying standard treatments

of Nash Equilibrium. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Auction Theory from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 23:

The proof follows from the definitions and properties established in 14.12. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

Solution approach: First recall the definitions and key theorems related to Signaling from 14.12. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 25:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Nash Equilibrium. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.